Multiscale Regression for Chronic Kidney Disease via Age-Curvature and Electrolytes

Mingyu Sun

Department of Mathematical & Computational Sciences,
University of Toronto Mississauga, Ontario L5L 1C6, Canada

e-mail: kimmich.sun@mail.utoronto.ca

Abstract - This study investigates the clinical modeling of chronic kidney disease (CKD) using generalized linear regression frameworks. A dataset of 400 patients from the UCI Machine Learning Repository was analyzed to evaluate the predictive value of serum sodium and potassium in classifying CKD status. Logistic regression results, supported by a Drop-in-Deviance test, indicated that including these electrolytes significantly improved model fit, with higher levels associated with reduced disease likelihood. Additionally, multivariable linear regression was applied to examine blood pressure as a function of age, serum creatinine, packed cell volume, blood urea, and hypertension status. A quadratic term for age was incorporated to model nonlinearity. The F-test confirmed a significant improvement in predictive performance, showing that blood pressure rises with age but at a slower rate in older individuals. These findings underscore the clinical and statistical relevance of incorporating serum-based biomarkers and age curvature in CKD risk modeling. The study offers interpretable and robust insights for enhancing disease stratification and may inform personalized approaches to early detection and management.

***Keywords*:** Chronic Kidney Disease; Clinical Predictors; Generalized Linear Models; Model Comparison

1 Introduction

Chronic kidney disease (CKD) is a progressive loss of renal function that affects a large portion of the global population and increases the risk of complications such as hypertension, anemia, and cardiovascular disease. Accurate and timely risk stratification is crucial for slowing disease progression and guiding clinical decision-making [1].

While numerous diagnostic markers exist, most prognostic models impose restrictive linear assumptions, overlooking key physiological interactions. Electrolyte imbalances and nonlinear age effects, as demonstrated by recent work on CKD progression [2], remain largely neglected in conventional frameworks despite their clear clinical relevance.

This study applies generalized linear modeling to a dataset of 400 patients from the UC Irvine Machine Learning Repository [3]. The analysis focuses on two critical aspects of CKD modeling: (1) assessing the incremental predictive value of incorporating serum sodium and potassium for disease classification, and (2) evaluating the potential nonlinear association between age and blood pressure that may require higher-order representation. By integrating key clinical features into statistically rigorous models, this work seeks to refine the precision and interpretability of CKD risk prediction, supporting the development of more personalized and clinically actionable decision tools.

2 Research Methodology

This study employs a retrospective analytical framework using the previously described dataset of 400 patient records from the UC Irvine Machine Learning Repository. Variable selection was guided by biomedical relevance, emphasizing indicators of renal function, electrolyte balance, and cardiovascular burden. Two statistical formulations were employed:

logistic regression for binary classification of CKD status, and multivariable linear regression for modeling blood pressure as a continuous outcome. To address potential nonlinear age effects, a polynomial specification was incorporated and assessed through nested model comparisons. Model evaluation relied on information-theoretic criteria and formal hypothesis testing.

2.1 Study Population and Variables

The dataset includes 24 clinically relevant variables spanning binary, continuous, and multicategorical formats. The primary outcome is a binary indicator denoting the presence or absence of chronic kidney disease at the time of data collection. Candidate predictors were selected based on clinical plausibility and diagnostic value. These included measures of renal function (e.g., serum creatinine, blood urea) [4], hematologic status (e.g., hemoglobin, packed cell volume), urinary characteristics (e.g., specific gravity, albumin levels), and comorbidities such as hypertension [5]. Blood pressure was also incorporated as a key physiological marker associated with both renal and cardiovascular risk.

3 Data Analysis

The analysis was performed using R Statistical Software 4.4.2 [6] to explore the predictive relationships between various clinical variables and the occurrence of chronic kidney disease. Binary logistic regression and multivariable linear regression were employed to evaluate potential predictors and model their significance. Additionally, the analysis included the use of diagnostic tests such as Goodness-of-Fit and Drop-in- Deviance tests to compare model performance and verify the significance of additional predictors.

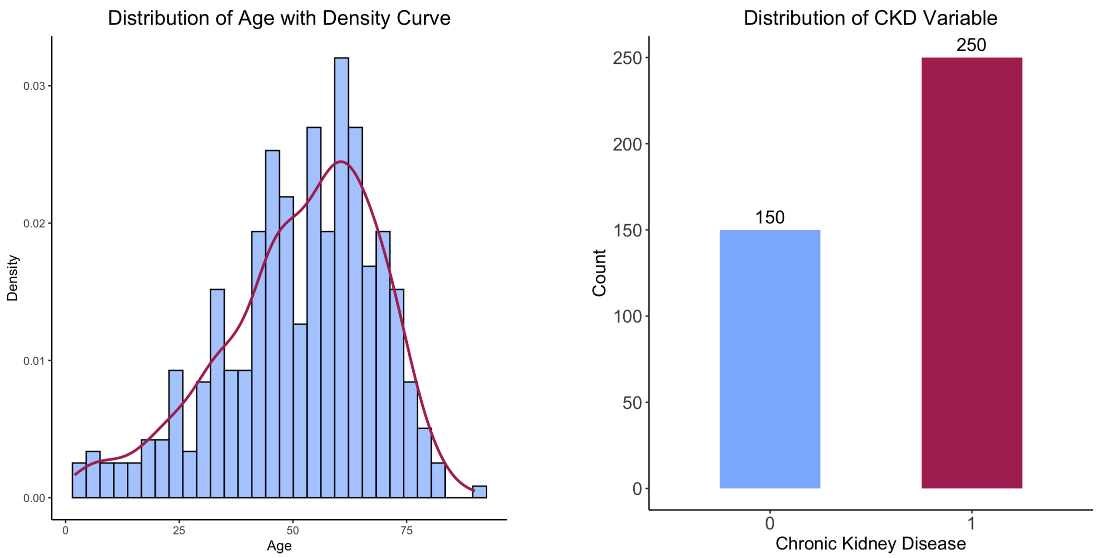
3.1 Exploratory Data Analysis

The dataset comprised a rich array of clinical and biochemical variables but included 242 observations with missing entries. To safeguard analytical rigor, variables with substantial missingness were excluded from exploratory analyses. Summary statistics for the retained variables are presented in Table 1 summarizes the retained variables as medians. CKD participants were markedly older (60.00 years vs. 46.50 years) and showed higher blood pressure and renal clearance markers, with blood urea nearly doubling (66.50 mg/dL vs. 33.00 mg/dL) and serum creatinine rising more than threefold (2.95 mg/dL vs. 0.90 mg/dL). Hemoglobin and packed cell volume were substantially lower (10.30 g/dL and 32.00%) than in non-CKD participants, consistent with CKD-related anemia [7]. Electrolyte shifts were limited: sodium declined slightly (136.00 mEq/L vs. 141.50 mEq/L), while potassium showed minimal change.

Variable	Unit	CKD Group	Non-CKD Group
Age	years	60.00 [49.75–67.00]	46.50 [34.75–58.00]
Blood Pressure	mm/Hg	80.00 [70.00–90.00]	70.00 [60.00–80.00]
Blood Urea	mg/dL	66.50 [40.00–111.50]	33.00 [23.75–44.00]
Hemoglobin	g/dL	10.30 [8.95–11.60]	15.00 [14.07–16.12]
Packed Cell Volume	%	32.00 [28.00–35.25]	46.00 [43.00–50.00]
Potassium	mEq/L	4.30 [3.90–4.90]	4.50 [3.70–4.90]
Serum Creatinine	mg/dL	2.95 [1.70–6.15]	0.90 [0.60–1.10]
Sodium	mEq/L	136.00 [131.75–139.00]	141.50 [138.00–146.00]

Table 1: Median and Interquartile Range for Key Clinical Variables in CKD vs Non-CKD Groups

Figure 1 complements these summaries by illustrating key distributions: Panel (a) depicts the age profile, revealing a right-shift among CKD cases, while Panel (b) shows the overall CKD vs. non-CKD composition, framing the dataset for subsequent modeling.



(a) (b)

Figure 1: Distribution of Age (a) and Chronic Kidney Disease Status (b)

3.2 Association between Electrolyte Levels and CKD Prediction

This section investigates whether the inclusion of serum sodium and potassium concentrations improves the predictive performance of a logistic regression model developed to classify individuals with or without chronic kidney disease. The initial model incorporates essential clinical variables such as age, blood pressure, serum creatinine, and blood urea, which serve as standard indicators in the clinical evaluation of renal function. Building on established CKD risk modeling frameworks that underscore the importance of integrating laboratory biomarkers into prognostic algorithms [8], a more comprehensive model is constructed by incorporating sodium and potassium concentrations as additional explanatory variables.

The analysis tests whether incorporating serum sodium and potassium significantly improves a baseline model of standard clinical predictors. The null hypothesis states that these electrolytes add no predictive value beyond conventional covariates, whereas the alternative posits that their inclusion yields a discernible and statistically meaningful enhancement in the model's ability to distinguish chronic kidney disease status.

Model	**AIC**	**Statistic**	**Age**	**Blood Pressure**	**Serum Creatinine**	**Blood Urea**	**Sodium**	**Potassium**
Reduced Model	178.6	β	0.0151	0.0429	5.8214	-0.0104	–	–
		p-value	0.3537	0.0444 *	1.93e-09 ***	0.5783	–	–
Full Model	165.2	β	0.0066	0.0500	5.7893	-0.0020	-0.2714	-0.5440
		p-value	0.7081	0.0464 *	1.3e-07 ***	0.9235	0.0001 ***	0.0062 **

Table 2: Results of Reduced and Full Binary Logistic Regression Models

The Drop-in-Deviance Test compared *Model* M_{reduced} with *Model* M_{full}, which incorporated sodium and potassium as additional predictors. The test yielded a p-value of $p = 4.2635 \times 10^{-6}$. Since this p-value is well below the 0.05 threshold, the null hypothesis that sodium and potassium do not contribute to the model is rejected. This result confirms that the full model provides a significantly better fit, as summarized in Table 2.

The positive coefficient of sodium indicates a significant negative relationship with the odds of getting chronic kidney disease. For every one unit increase in sodium, the odds of having chronic kidney disease decrease by approximately 23.77%, holding all other covariates constant. It suggests that individuals with higher sodium levels may have a lower likelihood of getting chronic kidney disease, potentially reflecting the importance of maintaining an appropriate balance of sodium levels in clinical assessments. Similarly, each additional unit of potassium level corresponds to a 41.96% decrease in the odds of developing chronic kidney disease while holding all other factors constant. This demonstrates a significant association between higher potassium levels and a reduced likelihood of chronic kidney disease. From a clinical perspective, this finding underscores the crucial role of potassium in the prevention and management of the disease. Specifically, low potassium levels might serve as an indicator of kidney dysfunction or related metabolic abnormalities.

3.3 Association between Nonlinear Age Effects and Blood Pressure Prediction

Age-related physiological changes often progress along nonlinear trajectories, creating inflection points that alter vascular and renal regulation [9]. Recognizing these nonlinear effects is clinically important for improving chronic kidney disease risk assessment and management [10]. This section examines whether including a quadratic age term (age^2) in a multivariable linear regression model improves its predictive performance for blood pressure. The base model includes age, serum creatinine, packed cell volume, blood urea, and hypertension status, selected for their established or suspected influence on blood pressure. To account for potential curvature, the squared age term (age^2) is incorporated as an additional predictor. Comparing the base and extended models evaluates whether modeling this nonlinear component enhances predictive accuracy and helps identify age thresholds where compensatory mechanisms shift, supporting earlier detection and more targeted interventions in chronic kidney disease populations.

To assess whether nonlinearities in age related physiological trajectories contribute meaningfully to blood pressure variation, a hypothesis testing framework was employed. The null hypothesis holds that the quadratic age term contributes no additional explanatory value beyond the linear specification. The alternative posits that its inclusion improves model performance by capturing subtle age related patterns not reflected in the linear form alone.

Characteristics	**Restricted Model**		**Unrestricted Model**	
	$\boldsymbol{\beta}$	p-value	$\boldsymbol{\beta}$	p-value
Age	0.0149	0.7673	0.6322	0.00351 **
Age^2	–	–	-0.0065	0.00340 **
Serum Creatinine	0.1805	0.2952	0.1667	0.32734
Packed Cell Volume	-0.2694	0.0294 *	-0.2809	0.02154 *
Blood Urea	-0.0046	0.8346	-0.0082	0.70793
Hypertension	5.4377	0.0106 *	5.6583	0.00717 **
AIC	2402.90		2396.10	

Table 3: Results of Restricted and Unrestricted Multivariable Linear Models

As summarized in Table 3, the analysis compared two multiple linear regression models, *Model* $M_{\text{restricted}}$ and *Model* $M_{\text{unrestricted}}$, to assess whether incorporating the quadratic age term (age^2) improved the model's predictive performance for blood pressure.

A goodness-of-fit test was performed to compare the two models by evaluating the residual sum of squares (RSS). The results of the goodness-of-fit test indicated an F-statistic of 8.7218 with a p-value of 0.003398, which is highly significant ($p < 0.01$). This result led to the rejection of the null hypothesis and acceptance of the alternative hypothesis, confirming that adding the quadratic age term significantly improves the model's predictive power for blood pressure.

The negative coefficient for the quadratic age term (age^2) suggests that the relationship between age and blood pressure is not purely linear. Instead, at older ages, the rate of increase in blood pressures slows down. This supports the inclusion of the quadratic term for age to better capture the nuanced age effects on blood pressure. And Additionally, both coefficients are statistically significant, confirming that age and its quadratic term play crucial roles in predicting blood pressure. For the factor hypertension, the coefficient of 5.6583 indicates that individuals with hypertension are predicted to have significantly higher blood pressure levels compared to those without hypertension, holding all other variables constant. This result, which is highly significant ($p < 0.01$), aligns with the well-established understanding that hypertension is a primary contributor to elevated blood pressure.

4 Conclusions and Limitations

From a clinical perspective, the inclusion of sodium and potassium in the model improves fit because these electrolytes play critical roles in kidney function. Imbalances in sodium and potassium levels are common indicators of renal impairment and are linked to fluid balance, blood pressure regulation, and metabolic processes [11]. Including these two variables helps capture the complex biochemical changes associated with chronic kidney disease, leading to a more accurate prediction model.

These finding has practical implications, especially for understanding the age acceleration effect in chronic kidney disease patients. The age acceleration effect represents the phenomenon where biological aging occurs at a faster rate than expected based on chronological age [12]. Among these patients, this can manifest as an increased risk of cardiovascular complications, reflected in altered blood pressure dynamics. The inclusion of a quadratic term for age in the model suggests that the relationship between age and blood pressure is not linear but exhibits complex, non-linear patterns. This insight can help tailor more effective interventions and monitoring strategies to manage blood pressure and reduce cardiovascular risk in chronic kidney disease patients.

Admittedly, this study is subject to several data-related limitations. First, the dataset contains missing values; despite mitigation efforts, imputation or exclusion may introduce bias. Second, the dataset is modest in size (400 observations), limiting statistical power and the detection of subtle associations. Third, survival or longitudinal data were unavailable, preventing time-to-event analyses and assessment of long-term prognostic value. Fourth, key covariates, including demographic and genetic factors, were not captured, potentially reducing model completeness. Finally, the study relies on a retrospective, single-source dataset, which may restrict generalizability to broader populations.

5 Acknowledgement

The author sincerely acknowledges that this study originates from work conducted in STA312 (Fall 2024) at the University of Toronto Mississauga under the guidance of Professor Nishan Mudalige, whose instruction and support were invaluable to the development of this research.

REFERENCES

[1] A. Marks *et al.*, "Looking to the future: Predicting renal replacement outcomes in a large community cohort with chronic kidney disease," *Nephrology Dialysis Transplantation*, vol. 30, no. 9, pp. 1507–1517, 2015. DOI: 10.1093/ndt/gfv080.

[2] H. Al-Wahsh *et al.*, "Investigating the relationship between age and kidney failure in adults with category 4 chronic kidney disease," *Canadian Journal of Kidney Health and Disease*, vol. 7, p. 2 054 358 120 966 819, Oct. 2020. DOI: 10.1177/2054358120966819.

[3] L. Rubini, P. Soundarapandian, and P. Eswaran, *Chronic kidney disease [dataset]*, 2015. DOI: 10.24432/C5G020.

[4] J. B. Echouffo-Tcheugui and A. P. Kengne, "Risk models to predict chronic kidney disease and its progression: A systematic review," *PLoS medicine*, vol. 9, no. 11, e1001344, 2012.

[5] N. Tangri *et al.*, "A dynamic predictive model for progression of ckd," *American Journal of Kidney Diseases*, vol. 69, no. 4, pp. 514–520, 2017.

[6] R Core Team, *R: A language and environment for statistical computing*, R Foundation for Statistical Computing, Vienna, Austria, 2024. [Online]. Available: https://www.R-project.org/.

[7] W. Pan, Y. Han, H. Hu, and Y. He, "Association between hemoglobin and chronic kidney disease progression: A secondary analysis of a prospective cohort study in japanese patients," *BMC Nephrology*, vol. 23, no. 1, p. 295, Aug. 2022. DOI: 10.1186/s12882-022-02920-6.

[8] N. Tangri, M. E. Grams, A. S. Levey, J. Coresh, *et al.*, "Multinational assessment of accuracy of equations for predicting risk of kidney failure: A meta-analysis," *JAMA*, vol. 315, no. 2, pp. 164–174, Jan. 2016. DOI: 10.1001/jama.2015.18202.

[9] C. P. Kovesdy *et al.*, "Age and outcomes associated with bp in patients with incident ckd," *Clinical Journal of the American Society of Nephrology*, vol. 11, no. 5, pp. 821–831, 2016.

[10] I. Ali, R. Chinnadurai, S. Ibrahim, *et al.*, "Adverse outcomes associated with rapid linear and non-linear patterns of chronic kidney disease progression," *BMC Nephrology*, vol. 22, p. 82, Feb. 2021. DOI: 10.1186/s12882-021-02282-5.

[11] C. P. Kovesdy *et al.*, "Serum potassium and adverse outcomes across the range of kidney function: A ckd prognosis consortium meta-analysis," *European Heart Journal*, vol. 39, no. 17, pp. 1535–1542, May 2018. DOI: 10.1093/eurheartj/ehy100.

[12] A. M. O'Hare *et al.*, "Age affects outcomes in chronic kidney disease," *Journal of the American Society of Nephrology*, vol. 18, no. 10, pp. 2758–2765, 2007.